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MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
Network Protocols & Architecture
LAB EXPERIMENT # 11

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Score: _____ Signature of the Lab Tutor: _____ Date: _____

Configuring PPPoE

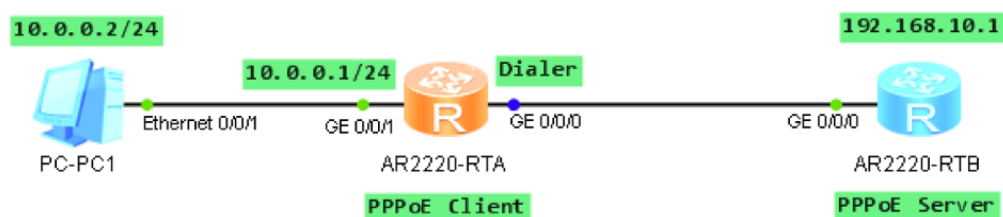
LAB PERFORMANCE INDICATOR	SUBJECT KNOWLEDGE	DATA ANALYSIS AND INTERPRETATION	ABILITY TO CONDUCT EXPERIMENT	SCORE
OBJECTIVE NO:				

Learning Objectives

As a result of this lab section, you should achieve the following tasks:

- ☐ Configure dialer interface for PPPoE and authentication of client over PPPoE
- ☐ Configuration of a dialer interface for PPPoE.
- ☐ Authentication of a client over PPPoE.

Topology



Scenario

The enterprise subscribes to a (typically high speed) DSL service from the service provider over which WAN services are supported. RTA is the enterprise edge router of different offices, and establish a connection to the service provider through the PPPoE server (RTB). The enterprise is required to establish a PPPoE dialer on the edge router to allow hosts in the local area network to access external resources transparently via the service provider network over PPPoE.

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Tasks

Step 1 Preparing the environment

```
<Huawei> system-view
Enter system view, return user view with Ctrl+Z.
[Huawei] sysname RTA

<Huawei> system-view
Enter system view, return user view with Ctrl+Z.
[Huawei] sysname RTB
```

Step 2 Configure PPPoE Server

The PPPoE server is not part of the enterprise network, however it is required to allow the enterprise edge router RTA to be authenticated. A PPPoE server dynamically allocates IP addresses to clients, and provides multiple authentication methods.

Configure the router to assign IP addresses to PPPoE clients. That is, configure a global address pool on the PPPoE server to dynamically allocate IP addresses to hosts.

```
[RTB] ip pool pool1
[RTB-ip-pool-pool1] network 192.168.10.0 mask 255.255.255.0
[RTB-ip-pool-pool1] gateway-list 192.168.10.1
[RTB-ip-pool-pool1] quit
```

Configure the system domain for PPPoE authentication. Create an authentication and authorization scheme and set the authentication and authorization mode to **local**.

```
[RTB] aaa
[RTB-aaa] authentication-scheme system
[RTB-aaa-authen-system] authentication-mode local
[RTB-aaa-authen-system] quit
[RTB-aaa] authorization-scheme system
[RTB-aaa-authen-system] authorization-mode local
[RTB-aaa-authen-system] quit
```

Create a domain and configure the authentication and authorization schemes of the domain.

```
[RTB-aaa] domain system
```

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```
[RTB-aaa-domain-system] authentication-scheme system
[RTB-aaa-domain-system] authorization-scheme system
[RTB-aaa-domain-system] quit
```

Configure a PPPoE client user name and password, that is create a local user.

```
[RTB-aaa] local-user user1@system password cipher huawei123
[RTB-aaa] local-user user1@system service-type ppp
[RTB-aaa] quit
```

Configure a virtual template, that is configure the PPPoE server to implement CHAP authentication on PPPoE clients and set the domain to **system**.

```
[RTB] interface virtual-template 1
[RTB-Virtual-Template1] ppp authentication-mode chap domain system
[RTB-Virtual-Template1] ip address 192.168.10.1 255.255.255.0
[RTB-Virtual-Template1] remote address pool pool1
[RTB-Virtual-Template1] quit
```

Enable the PPPoE server function on Interface, that is, bind the virtual template to interface.

```
[RTB] interface GigabitEthernet 0/0/0
[RTB-GigabitEthernet0/0/0] pppoe-server bind virtual-template 1
[RTB-GigabitEthernet0/0/0] quit
```

Step 3 Configure PPPoE Client

Create and configure a dialer interface and dialer access control list. The number of the list must be the same as that of dialer-group 1.

```
[RTA] dialer-rule
[RTA-dialer-rule] dialer-rule 1 ip permit
[RTA-dialer-rule] quit
[RTA] interface dialer 1
[RTA-Dialer1] dialer user user1
[RTA-Dialer1] dialer-group 1 [RTA-Dialer1]
dialer bundle 1
```

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Configure the same user name and password as those configured on the PPPoE server for **CHAP** authentication. Configure the dialer interface to obtain an IP address from the remote device through PPP negotiation.

Using the **dialer queue-length** command, you can set the buffer queue length on a dial interface. The number **8** indicates the number of packets allowed in the buffer queue.

```
[RTA-Dialer1] ppp chap user user1@system
[RTA-Dialer1] ppp chap password cipher huawei123
[RTA-Dialer1] ip address ppp-negotiate
[RTA-Dialer1] dialer timer idle 300 [RTA-Dialer1]
dialer queue-length 8
[RTA-Dialer1] quit

[RTA] interface GigabitEthernet 0/0/0
[RTA-GigabitEthernet0/0/0] pppoe-client dial-bundle-number 1
[RTA-GigabitEthernet0/0/0] quit
[RTA] interface GigabitEthernet 0/0/1
[RTA-GigabitEthernet0/0/0] ip address 10.0.0.1 24
[RTA-GigabitEthernet0/0/0] quit
```

(Optional) --- Configure **NAT** on the device to translate LAN users' private IP addresses into public IP addresses so that the users can dial up to the Internet.

```
[RTA] acl number 3002
[RTA-acl-adv-3002] rule 5 permit ip source 192.168.10.0 0.0.0.255
[RTA-acl-adv-3002] rule 10 permit ip source 10.0.0.0 0.0.0.255
[RTA-acl-adv-3002] quit

[RTA] interface dialer 1
[RTA-Dialer1] nat outbound 3002
[RTA-Dialer1] quit
```

Configure a static route to the PPPoE server. Set the outbound interface to **Dialer1**.

```
[RTA] ip route-static 0.0.0.0 0 dialer 1
[RTA] quit
```

Sample output of Wireshark on RTA GigabitEthernet 0/0/0 interface.

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Configure the PC to have static IP.

No.	Time	Source	Destination	Protocol	Info
1	0.000000	HuaweiTe_d2:3e:4f	Broadcast	PPPoED	Active Discovery Initiation (PADI)
2	0.094000	HuaweiTe_c5:80:78	HuaweiTe_d2:3e:4f	PPPoED	Active Discovery offer (PADO) AC-Name='RTB00e'
3	0.141000	HuaweiTe_d2:3e:4f	HuaweiTe_c5:80:78	PPPoED	Active Discovery Request (PADR) AC-Name='RTB00e'
4	0.219000	HuaweiTe_c5:80:78	HuaweiTe_d2:3e:4f	PPPoED	Active Discovery Session-confirmation (PADS) AC-Name='RTB00e'
5	0.219000	HuaweiTe_c5:80:78	HuaweiTe_d2:3e:4f	PPP LCP	Configuration Request

Host Name:

MAC Address:

IPv4 Configuration

☒ Static ☐ DHCP ☐ Obtain DNS server address automatically

IP Address: DNS1:

Subnet Mask: DNS2:

Gateway:

IPv6 Configuration

☒ Static ☐ DHCPv6

IPv6 Address:

Prefix Length:

IPv6 Gateway:

Verify the connectivity with PPPoE server.

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```
PC> ping 192.168.10.1
```

```
Ping 192.168.10.1: 32 data bytes, Press Ctrl_C to break
From 192.168.10.1: bytes=32 seq=1 ttl=254 time=62 ms From
192.168.10.1: bytes=32 seq=2 ttl=254 time=32 ms From
192.168.10.1: bytes=32 seq=3 ttl=254 time=46 ms From
192.168.10.1: bytes=32 seq=4 ttl=254 time=16 ms From
192.168.10.1: bytes=32 seq=5 ttl=254 time=31 ms
```

```
--- 192.168.10.1 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
0.00% packet loss
 round-trip min/avg/max = 16/37/62 ms
```

The **display nat outbound** command displays information about outbound NAT.

```
[RTA] display nat outbound
```

```
NAT Outbound Information:
```

```
-----
Interface                Acl      Address-group/IP/Interface      Type      -----
-----
Dialer1                   3002     192.168.10.254                 easyip
-----
Total : 1
```

Verify the configuration to check the PPPoE session status. The configuration is correct if the output shows PPPoE session status **UP**.

```
[RTA] display pppoe-client session summary
```

```
PPPoE Client Session:
```

ID	Bundle	Dialer	Intf	Client-MAC	Server-MAC	State
1	1	1	GE0/0/0	00e0fc4b63af	00e0fcd37a6e	UP

To display brief information, including the IP address, subnet mask, physical status, link-layer protocol status, and number of interfaces in different states.

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```
[RTA] display ip interface brief
```

```
*down: administratively down
```

```
^down: standby
```

```
(l): loopback
```

```
(s): spoofing
```

```
The number of interface that is UP in Physical is 3
```

```
The number of interface that is DOWN in Physical is 2
```

```
The number of interface that is UP in Protocol is 2
```

```
The number of interface that is DOWN in Protocol is 3
```

Interface	IP Address/Mask	Physical	Protocol
Dialer1	192.168.10.254/32	up	up(s)
GigabitEthernet0/0/0	unassigned	up	down
GigabitEthernet0/0/1	unassigned	down	down
GigabitEthernet0/0/2	unassigned	down	down
NULL0	unassigned	up	up(s)

Verify the configurations on PPPoE server by checking the session status.

```
[RTB] display pppoe-server session all
```

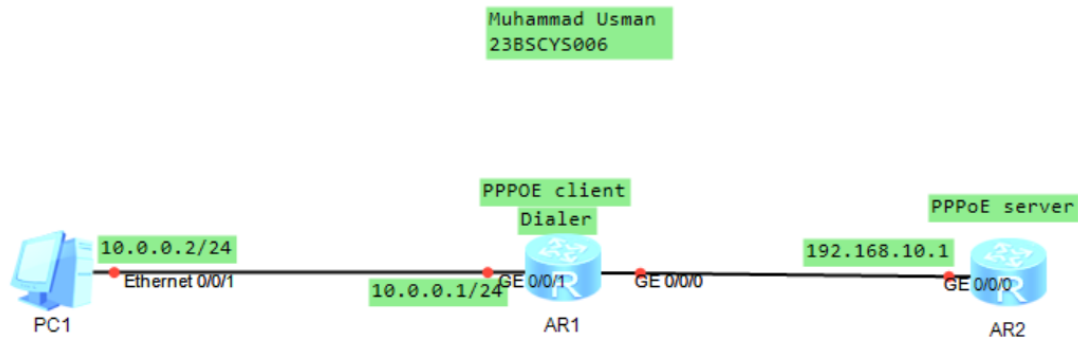
SID	Intf	State	OIntf	RemMAC	LocMAC
1	Virtual-Template1:0	UP	GE0/0/0	00e0.fc4b.63af	00e0.fcd3.7a6e

Tasks

Try a different topology and demonstrate your learning.

Upload the **.topo** **.efz** **.cfg** files

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```

AR1
AR1  AR2
*down: administratively down
^down: standby
(l): loopback
(s): spoofing
The number of interface that is UP in Physical is 4
The number of interface that is DOWN in Physical is 1
The number of interface that is UP in Protocol is 3
The number of interface that is DOWN in Protocol is 2

Interface                IP Address/Mask      Physical  Protocol
Dialer1                  192.168.10.254/32    up        up(s)
GigabitEthernet0/0/0     unassigned            up        down
GigabitEthernet0/0/1     10.0.0.1/24          up        up
GigabitEthernet0/0/2     unassigned            down      down
NULL0                    unassigned            up        up(s)

[23BSCYS006-R1]dis ppp
[23BSCYS006-R1]dis pppoe
[23BSCYS006-R1]dis pppoe-server
[23BSCYS006-R1]dis pppoe-client ?
  session Display PPPoE Client session information
[23BSCYS006-R1]dis pppoe-client se
[23BSCYS006-R1]dis pppoe-client session ?
  packet Packet/Byte count information
  summary Session summary information
[23BSCYS006-R1]dis pppoe-client session s
[23BSCYS006-R1]dis pppoe-client session summary
PPPoE Client Session:
ID  Bundle  Dialer  Intf          Client-MAC  Server-MAC  State
1   1        1       GE0/0/0       00e0fcd21f00 00e0fc753bb3 UP
[23BSCYS006-R1]

<23BSCYS006-R2>system-view
Enter system view, return user view with Ctrl+Z.
[23BSCYS006-R2]dis ip int br
*down: administratively down
^down: standby
(l): loopback
(s): spoofing
The number of interface that is UP in Physical is 3
The number of interface that is DOWN in Physical is 2
The number of interface that is UP in Protocol is 2
The number of interface that is DOWN in Protocol is 3

Interface                IP Address/Mask      Physical  Protocol
GigabitEthernet0/0/0     unassigned            up        down
GigabitEthernet0/0/1     unassigned            down      down
GigabitEthernet0/0/2     unassigned            down      down
NULL0                    unassigned            up        up(s)
Virtual-Templatel        192.168.10.1/24      up        up

[23BSCYS006-R2]dis pppoe
[23BSCYS006-R2]dis pppoe-client
[23BSCYS006-R2]dis pppoe-server se
[23BSCYS006-R2]dis pppoe-server session a
[23BSCYS006-R2]dis pppoe-server session all
SID Intf          State OIntf          RemMAC          LocMAC
1   Virtual-Templatel:0  UP   GE0/0/0          00e0.fcd2.1f00 00e0.fc75.3bb3
[23BSCYS006-R2]
  
```


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143	330.016000	HuaweiTe_d2:1f:00	HuaweiTe_75:3b:b3	PPP LCP	60 Echo Request
144	330.031000	HuaweiTe_75:3b:b3	HuaweiTe_d2:1f:00	PPP LCP	60 Echo Request
145	330.031000	HuaweiTe_75:3b:b3	HuaweiTe_d2:1f:00	PPP LCP	60 Echo Reply
146	330.031000	HuaweiTe_d2:1f:00	HuaweiTe_75:3b:b3	PPP LCP	60 Echo Reply
147	340.016000	HuaweiTe_d2:1f:00	HuaweiTe_75:3b:b3	PPP LCP	60 Echo Request
148	340.031000	HuaweiTe_75:3b:b3	HuaweiTe_d2:1f:00	PPP LCP	60 Echo Request
149	340.031000	HuaweiTe_75:3b:b3	HuaweiTe_d2:1f:00	PPP LCP	60 Echo Reply
150	340.031000	HuaweiTe_d2:1f:00	HuaweiTe_75:3b:b3	PPP LCP	60 Echo Reply

```
> Ethernet II, Src: HuaweiTe_d2:1f:00 (00:e0:fc:d2:1f:00), Dst: HuaweiTe_75:3b:b3 (00:e0:fc:75:3b:b3)
✓ PPP-over-Ethernet Session
    0001 .... = Version: 1
    .... 0001 = Type: 1
    Code: Session Data (0x00)
    Session ID: 0x0001
    Payload Length: 10
✓ Point-to-Point Protocol
    Protocol: Link Control Protocol (0xc021)
✓ PPP Link Control Protocol
    Code: Echo Request (9)
    Identifier: 8 (0x08)
    Length: 8
    Magic Number: 0x0e76eab5
```

```
PC1
Basic Config Command MCPacket UdpPacket Console
From 192.168.10.1: bytes=32 seq=3 ttl=254 time=15 ms
From 192.168.10.1: bytes=32 seq=4 ttl=254 time=32 ms
From 192.168.10.1: bytes=32 seq=5 ttl=254 time=15 ms

--- 192.168.10.1 ping statistics ---
 5 packet(s) transmitted
 4 packet(s) received
20.00% packet loss
round-trip min/avg/max = 0/23/32 ms

PC>ping 192.168.10.1

Ping 192.168.10.1: 32 data bytes, Press Ctrl_C to break
From 192.168.10.1: bytes=32 seq=1 ttl=254 time=32 ms
From 192.168.10.1: bytes=32 seq=2 ttl=254 time=15 ms
From 192.168.10.1: bytes=32 seq=3 ttl=254 time=16 ms
From 192.168.10.1: bytes=32 seq=4 ttl=254 time=31 ms
From 192.168.10.1: bytes=32 seq=5 ttl=254 time=31 ms

--- 192.168.10.1 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
round-trip min/avg/max = 15/25/32 ms

PC>
```